1.4.2 Molar Mass:

Molar mass is defined as the mass of one mole of a substance. The molar mass of a compound is equal to the sum of the relative atomic masses of its constituents expressed in g mol^{-1}.

Examples:

- relative atomic mass of one hydrogen atom = 1.008 u
- molar mass of hydrogen atom = 1.008 g mol^{-1}
- relative molecular mass of glucose = 180 u
- molar mass of glucose = 180 g mol^{-1}

1.4.3 Molar Volume:

The volume occupied by one mole of any substance in the gaseous state at a given temperature and pressure is called molar volume.

Conditions	Volume occupied by one mole of any gaseous substances (in litre)
273 K and 1 bar pressure (STP)	22.71
273 K and 1 atm pressure	22.4
298 K and 1 atm pressure (Room Temperature & pressure (SATP)	24.47

Evaluate Yourself

3a) Calculate the number of moles present in 9 g of ethane.

3b) Calculate the number of molecules of oxygen gas that occupies a volume of 224 ml at 273 K and 3 atm pressure.

1.5 Gram Equivalent Concept:

Similar to mole concept gram equivalent concept is also widely used in chemistry especially in analytical chemistry. In the previous section, we have understood that mole concept is based on molecular mass. Similarly gram equivalent concept is based on equivalent mass.

Definition:

Gram equivalent mass is defined as the mass of an element (compound or ion) that combines or displaces 1.008 g hydrogen or 8 g oxygen or 35.5 g chlorine.

Consider the following reaction:

$$Zn + H_2SO_4 \rightarrow ZnSO_4 + H_2$$

In this reaction 1 mole of zinc (i.e. 65.38 g) displaces one mole of hydrogen molecule (2.016 g).

Mass of zinc required to displace 1.008 g hydrogen is

$$= \frac{65.38}{2.016} \times 1.008$$

$$= \frac{65.38}{2}$$

The equivalent mass of zinc = 32.69

The gram equivalent mass of zinc
= 32.69 g eq^{-1}

Equivalent mass has no unit but gram equivalent mass has the unit g eq^{-1}

It is not always possible to apply the above mentioned definition which is

11th Std Chemistry 001-037.indd 8 06-01-2021 15:02:43